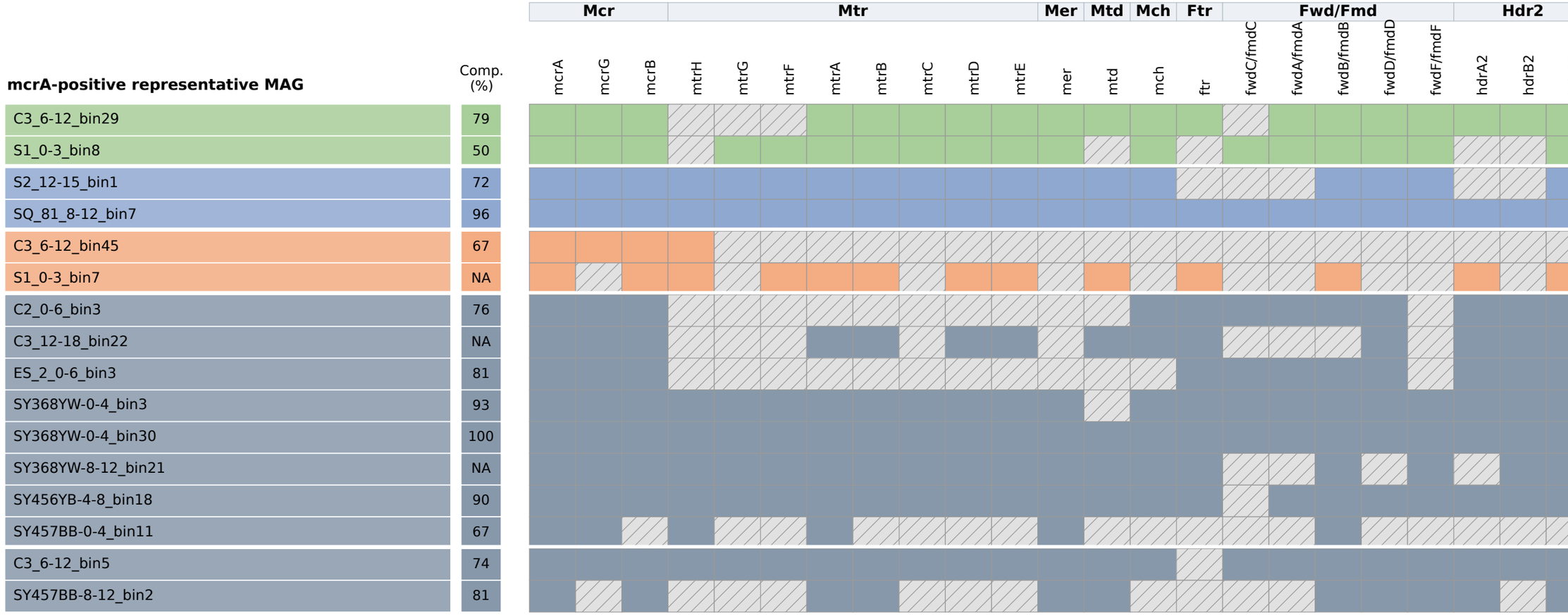
A. Reference-style MAG gene-detection matrix (no tree)

Conceptual $CH_4 \rightarrow CO_2$ carbon pathway; HdrA2/B2/C2 shown separately as an electron-transfer/cofactor-recycling group.



Filled cells: genomic detection, not abundance. All 16 hosts: GTDB Halobacteriota; 3 supplemental MAGs use representative-proxy taxonomy. No operon boundaries inferred.